



## Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

### DEPARTMENT OF CSE

#### PROJECT ABSTRACT SUBMISSION

#### COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

A.Y. 2026 - 2027

| TITLE OF THE PROJECT | ReLoop AI – Agentic E-Waste Recovery & Circularity Orchestrator |                    |
|----------------------|-----------------------------------------------------------------|--------------------|
| S. No.               | University ID                                                   | Name               |
| 1                    | 2420030734                                                      | N Sai Suhaas       |
| 2                    | 2420030641                                                      | A Sudarsan Krishna |
| 3                    | 2420030769                                                      | K Sai Dhanush      |
| Name of the Guide    |                                                                 | K Venkateswari     |

#### Abstract

Electronic waste (e-waste) is rapidly increasing, while many discarded electronic devices still contain reusable components or have potential for repair, refurbishment, or reuse. However, existing e-waste management processes often depend on manual assessment and mainly focus on collection, classification, and recycling. This can lead to unnecessary disposal of usable devices, loss of valuable materials, and inefficient coordination between users, collection centres, repair facilities, refurbishers, and recyclers. Therefore, an intelligent system is needed to identify the most suitable recovery option for each device and improve the overall e-waste recovery process.

The proposed ReLoop AI – Agentic E-Waste Recovery & Circularity Orchestrator is an Agentic AI-based application that automates the recovery journey of electronic devices. The system will analyze device details, condition, faults, and available diagnostic information using AI agents to assess its recovery potential. A circularity decision agent will determine whether the device should be repaired, reused, refurbished, recovered for parts, or recycled. The system will then use an intelligent partner-matching agent to identify suitable recovery partners and coordinate pickup, processing, status tracking, and notifications, reducing manual intervention throughout the process.

The application will be implemented using a Service-Oriented Architecture (SOA) and microservices approach, with services for user management, device management, AI assessment, circularity decisions, partner matching, recovery coordination, and notifications. REST APIs, API Gateway, Eureka Service Registry, JWT-based authentication, role-based authorization, and Kafka-based event-driven communication will be integrated to provide a scalable and secure architecture. The project aims to transform e-waste management from a simple collection-and-disposal process into an intelligent circular recovery ecosystem, increasing reuse and recovery while reducing unnecessary electronic waste. The project aligns with SDG 12: Responsible Consumption and Production and will result in a practical, scalable, hackathon-oriented application demonstrating the integration of Agentic AI, multiple AI tools, and SOA-based microservices.

Project Architecture/Project Overview:

